Statement of Objectives

My primary research interests lie in the intersection of machine learning (ML) and computational biology. How can we explore the biomolecular footprints of variation and heredity, combining statistical analysis with experimental data? Can this combination of information and medicine cause medical breakthroughs in vaccine and antibody development to make healthcare more affordable? These questions inspired me to pursue my current research interest in protein-protein interaction prediction using multi-modal properties and multitask fine-tuning. At Texas A&M University (TAMU), my research projects were focused on physiological data interpretation in healthcare. My main goal for graduate school will be to understand the unique biomolecules using available medical data, statistical analysis, and physical modeling. My research and academic experiences have led to multiple publications and presentations, with several awards and *an NSF CSGrad4US fellowship fully funding the first three years of my graduate school*.

Research Experience in ML and Computational Biology. Proteins are fundamental biomolecules that regulate and conduct nearly every necessary physiological action. From the individual 1D amino acid sequence, proteins fold into 3D structures and interact with each other to function. Despite the revolutionary breakthrough in protein 3D structure prediction, inter-protein structure prediction remains unsolved and needs more innovation. Motivated by this, my undergraduate thesis project was on *developing a machine learning algorithm that predicts inter-protein interactions using multi-modal data and auxiliary tasks* under the supervision of Dr. Yang Shen. This project provided me with a unique opportunity to work on an open-ended research problem from the ground up. I brought structure to the project, which required understanding the existing literature on protein structure, the nature of its interaction, and available tools by identifying the core areas of improvement. While most of the recent works focused on complicated machine learning models [1], we recognized a way to improve performance by maximum utilization of valid features and omitting noises and redundancy. We presented a new approach to improve inter-protein contact prediction via incorporating multi-modal data synergized with auxiliary predictive tasks. When I was researching machine learning methods to use the abundance of protein sequences, I realized that we can think of proteins like a human language. The way a couple of letters form words and eventually sentences, a very similar relationship can be established among amino acids to protein sequences and their 3D folding. Thus, successful tools from NLP, like pre-trained BERT [2] and HRNN [3] can be used as protein feature embedders. I was able to prove a gradual improvement by incorporating multi-modal synergy. Out of 75 projects, my thesis was selected to represent the electrical engineering department and won the second prize from the TAMU engineering board members.

After graduation, I joined Texas Instruments as a test and product engineer. Side by side, I continued my research collaboration with Dr. Yang Shen to publish my work. During this work, I found that one of the most important problems is balancing the pool of consequence data with limited causal data and this problem can be solved with complementary features which can enhance the interactive correlation. I decided to reduce the amount of data and introduced additional evolutionary information through multiple sequence alignment. I also faced the bottleneck problem of GPU memory and engineered my model accordingly. This work led to *Does Inter-Protein Contact Prediction Benefit from Multi-Modal Data and Auxiliary Tasks?,* accepted to NeurIPS workshop MLSB'22 [4]. As the lead of the project, I identified knowledge gaps in the current work, defined and scoped the research problem, analyzed the workload data, implemented the deep learning models, ran the experiments, discussed the results, and wrote the paper.

Other Research Experience. My research journey began under Dr. Farzan Sasangohar at the Applied Cognitive Ergonomics Lab, with a focus on remote health monitoring with real-time physiological data. In this project, *we developed an Android smartwatch app to assess post-traumatic stress disorder (PTSD)* episodes and proactively support veterans through wearable smart sensing technology. PTSD is a mental health disorder especially prevalent among veterans. PTSD attacks are detected through real-time heart rate and pulse monitoring [5]. While designing the app, I initiated to interview several PTSD patients. Based on their feedback and evaluation, I developed a "breathing exercise" feature to detect hyper-arousal triggers and provide instant support to resolve them through breathing and refocus exercises. I was proud to present my work as the only sophomore in the Gulf Coast Undergraduate Research Symposium (GCURS'18) at Rice University. As I used physical sensor data (e.g. heart rate) to detect PTSD, I got curious about these sensors' electrical mechanisms and signal analysis features.

My Research Experience for Undergrads (REU) project on *Wearable Device Development to Measure Continuous Blood Pressure (BP)* helped me learn both 1) the electrical side of sensor development and 2) software development to extract important morphologies. Working under the supervision of Dr. Roozbeh Jafari, we set out to address the challenge of developing a cuffless continuous BP monitoring system. The current BP monitoring methods depend on a bulky and obtrusive cuff that only allows sporadic blood pressure readings. Continuous BP readings would allow early detection of cardiovascular disease, reducing the risk of a sudden heart attack. To enable continuous BP monitoring, we presented an approach to use bio-impedance (Bio-Z) signals from an arterial pulse using noninvasive sensors that can be integrated into wearable devices [6]. While working with a single current source, I realized that interference between two Bio-Z sensors can result in noisy data and inaccurate pulse transit time. My main aim was to improve Bio-Z signal quality using separate current sources for each Bio-Z voltage channel, with different current injection frequencies for each current source. I have designed and conducted over 50 experiments to collect data from human subjects. As I observed the data, I started to draw conclusions and design experiments based on the feedback. At last, analyzing the collected data from four different subjects, I was able to prove and support that the multi-frequency current injection can capture better Bio-Z signals and thus result in a more accurate BP reading. This work led to my first co-authored paper, *Multi-source Multi-frequency Bio-impedance Measurement Method for Localized Pulse Wave Monitoring* (EMBC'20). I also presented my work at the GCURS'20 and won the "Excellence in Research " award. I deeply enjoyed the arterial blood flow simulation and signal processing part of this project. Thus, I elected to participate in the Undergraduate Research Scholar program as part of the Engineering Honors and selected bioinformatics as my thesis topic, which is heavily focused on biological model development and data analysis.

Mentoring and Volunteering. Attending higher education is a privilege and offers a great platform to interact with young students. My other goal in graduate school is to mentor and support young female students from high school and college to help them develop technical skills beyond the classroom. Throughout my college, I participated in mentorship and volunteering programs. To make educational content more accessible in my native language, I volunteered at Khan Academy as a Bengali translator. I also worked as a peer mentor for introductory engineering and physics courses.

Future Research Interests. As a graduate student, I plan to build computational models that can generate diverse and accurate representations of biomolecular structures and associations. I plan to extend my work on inter-protein interaction by developing ML-based tools to study biomolecular interactions and complexes responsible for particular phenotypes including diseases, like cancer, Alzheimer's, diabetes, and schizophrenia. One of the examples is gene mapping from different species to the same feature space to correlate complex synthetic lethal interactions. Columbia having one of the best computer science and computational biology programs in the country will provide the opportunities to find a well-structured problem and solve it through collaborations with scientists from computer science, genetics, psychology, and medicine – this tight integration is a unique strength. Specifically, I would be interested in working with **Dr. David A. Knowles** and **Dr. Itsik Pe'er**. One of Dr. Knowles's recent papers on predicting multiple genetic marks jointly by combining mutagenesis with Mendelian randomization [7] is very relevant to my research interest. Since I have analyzed the impact of multi-task learning in inter-protein contact prediction, this domain is very close to my past research work. Additionally, genomic marks play extremely important roles in gene regulation; that is why similar exploration would provide a strong background for my research interest in the robust modeling of expression-specific genomic causation. I also find Dr. Pe'er's work very inspiring. One of his recent works addresses the problem of generating synthetic longitudinal multinode data from microbial environments [8]. Though this work does not use ML techniques, it uses intensive biological simulation to generate the most important aspect of successful ML models which is noiseless data. The lack of noiseless, unbiased data is a major limitation to determine genotype-phenotype correlation as well. Extending my work under his supervision would give me strong support for datasets and necessary tool development in order to explore expression-specific genomic modeling. Moreover, I would be open to work with any advisor aligned with my research interest in ML and computational biology.

Future Career Goal. As a first-generation immigrant woman raised by a single mother from a low-income family, my journey was not easy. All my experiences have enriched my vision and aim to give back to society through research and innovation. After graduate studies, I want to build a career in research and want my work to be known as trustworthy, innovative, and impactful.

Reference

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